

Ace is on the case again, tracking down some penguin smugglers. First order of business is to get his suspect list in order so he can contact them and setup meetings. Help Ace get organized!



Input

Your input will be a list of e-mail addresses. There may be more than one per line separated by spaces or commas. `;;; on a line by itself ends input.`

```
abc@Example.com xyz@example.net
123@example.edu, codewarsisgreat@hpe.com abc@XYZ.com
heretherebeDRAGONS@hpe.com
abc@example.com
;;;
```

Output

Collect the e-mail addresses and eliminate the duplicates (ignore case: `abc@xyz.com` should be counted the same as `ABC@xYZ.CoM`). Any address you collect as unique should be converted to lowercase. Once you are done eliminating duplicates, sort the addresses in Ascending alphabetical order, and output them to the screen in Outlook separated (semi-colon) format, one address per line:

```
123@example.edu;
abc@example.com;
abc@xyz.com;
codewarsisgreat@hpe.com;
heretherebedragons@hpe.com;
xyz@example.net;
```

Discussion

Reminder: have you run your solution against **all** of the student data sets?

Additional Example

Input 2	Output 2
<pre>a@a.com a@a.com a@a.com a@a.com a@a.com A@A.com A@A.com a@a.com a@a.com a@a.com a@a.com, A@A.com a@a.com, a@a.com, a@a.com a@a.com a@a.com a@b.com ;;;</pre>	<pre>a@a.com; a@b.com;</pre>